

Design Proposal for MZIs and Ring Resonators

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Abstract—We propose a silicon photonic test chip, built on 500×220 nm strip waveguides, containing unbalanced Mach-Zehnder interferometers with varying path-length differences to extract the waveguide group index, and add-drop ring resonators with varying materials. Device performance is simulated using Lumerical MODE and INTERCONNECT, and a grating-coupler loopback structure is included to calibrate the measured spectra.

Index Terms—Mach-Zehnder Interferometer, Ring Resonator

I. INTRODUCTION

A Mach-Zehnder Interferometer (MZI) is a 2-port device which first splits an incoming beam into two via a Y-branch and then combines them again via another. The two Y-branches are connected with waveguides of varying lengths such that the output of the interference pattern is dependent on the path length difference of the two waveguides and their effective indices.

In this short project proposal, we look at five MZIs with path length differences $\Delta L = 25 \mu\text{m}, 50 \mu\text{m}, 100 \mu\text{m}, 200 \mu\text{m},$ and $250 \mu\text{m}$ to illustrate the above phenomena. In addition to the MZIs, we have also added grating couplers and Y-branches for input and output. For our purposes, we have also added two ring resonators—one made of silicon and the other made of SiN—with bus waveguides.

II. MODELING, SIMULATIONS, AND THEORY

A. Modeling Waveguides

To develop MZIs, we use standard SOI (silicon-on-insulator) strip waveguides. The waveguides have dimensions of $500 \text{ nm} \times 220 \text{ nm}$. The 220 nm width is an industry standard for most silicon based foundries. We will be analyzing our MZIs at the standard telecom wavelength of 1550 nm . As can be seen in Figs. 1 (a) and (b), the refractive indices of Si and SiO_2 are 3.4755 and 1.444 , respectively at $1.55 \mu\text{m}$.

We now use the above values to simulate our waveguide geometry in Lumerical MODE solver. We perform an eigen-mode simulation to analyze the TE and TM modes. Figs. 2 (a) and (b) the electric field intensity profiles of the fundamental TE and TM modes, respectively. The effective index of the TE mode is $2.44 + i1.25 \times 10^{-9}$ with a loss of $4.4 \times 10^{-4} \text{ dB/cm}$. The mode is contained within an effective area of $0.19 \mu\text{m}^2$. On the other hand, the effective index of the TM mode is $1.76 + i8.50 \times 10^{-10}$ with a loss of $2.9 \times 10^{-4} \text{ dB/cm}$ and an effective area of $0.37 \mu\text{m}^2$. However, the TE polarization

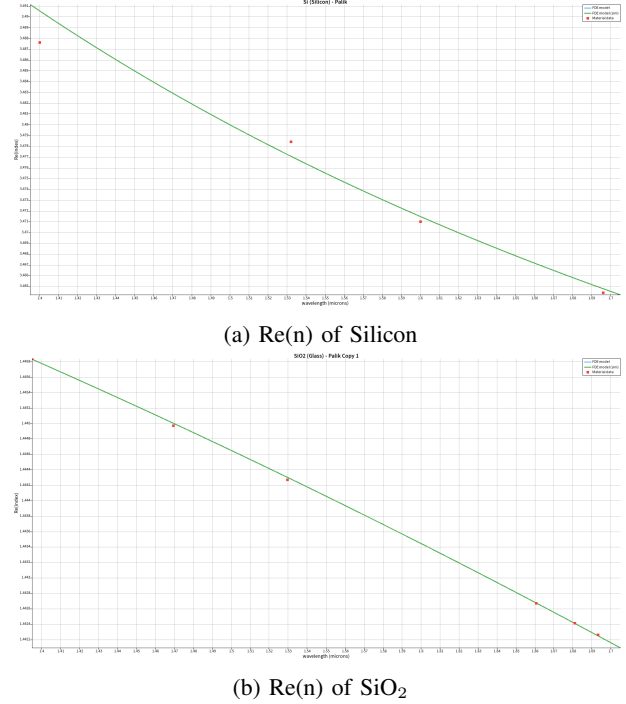


Fig. 1: Refractive index (n) properties of silicon and SiO_2 for a standard SOI strip waveguide. Note, we only highlight the real part of the indices as the imaginary parts are negligibly small.

fraction (E_x) for the former is 98 and 4 for the latter indicating that the fundamental mode in the waveguide is quasi-TE.

Next, we analyze the effective and group indices as a function of wavelengths. As can be seen in Figs. 3 (a) and (b), the real part of the effective index value is 2.4486 and the group index is 4.2038 for the quasi-TE mode at $1.55 \mu\text{m}$.

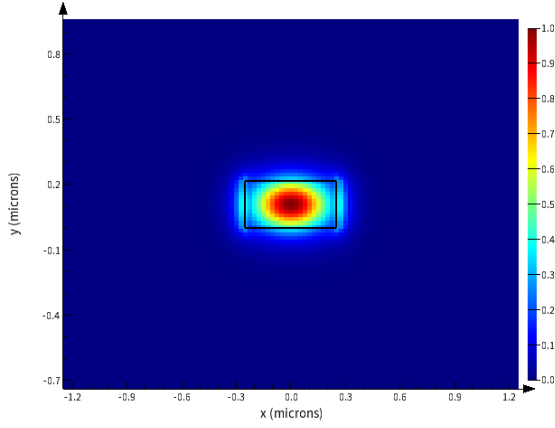
Lastly, we exported the n_{eff} vs λ data to MATLAB to develop a compact model. We curve fit n_{eff} data to the polynomial

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

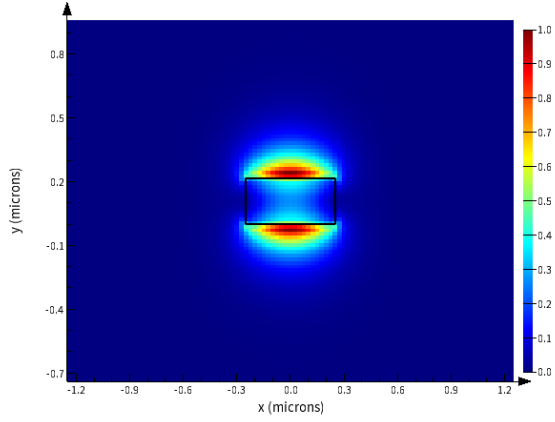
and extract the values of n_1 , n_2 , and n_3 . With $\lambda_0 = 1.55 \mu\text{m}$ as the central wavelength, we get $n_1 = 2.44$, $n_2 = -1.13$, and $n_3 = -0.044$.

B. Mach-Zehnder Interferometer Theory

We now begin to analyze our Mach-Zehnder Interferometer. In free space or fiber optics, an MZI is simply composed



(a) $|E|^2$ intensity profile for a quasi-TE mode



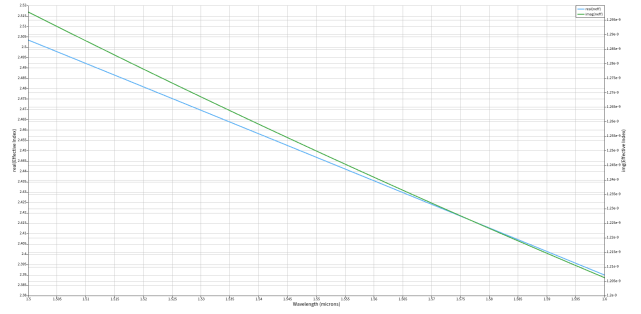
(b) $|E|^2$ intensity profile for a quasi-TM mode

Fig. 2: Intensity profiles for the fundamental TE and TM modes

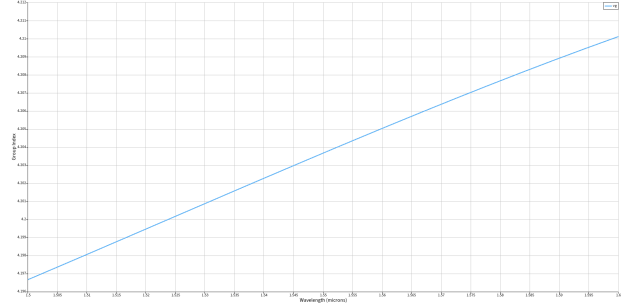
of two beam splitters and mirrors such that the first beam-splitter splits the input beam into two, the mirrors then guide them to the second beam-splitter, which further combines the two beams. For a balanced interferometer with 50:50 beam splitters, equal arms, and index, the output constructively interferes in only one output port of the second beam-splitter and destructively interferes in the other port. If we introduce a phase shift in one of the arms, the MZI becomes imbalanced. This can either be due to the incorporation of a material with a different refractive index (a LiNbO_3 material with a $\chi^{(2)}$ nonlinearity, for example) or by simply introducing additional optical path length. In this proposal, we will explore the latter.

In photonic circuits, the above setup can be implemented by replacing the beam-splitters and mirrors with Y-branches and silicon waveguides, respectively. Additionally, the input and output is observed via grating couplers. These elements are provided in the SiEPIC process design kit (PDK).

We first begin by analyzing the MZI theoretically. For an input field with electric field amplitude E_i and intensity $I_i \propto |E_i|^2$, the input Y-branch splitter splits the field into



(a) Effective index for the quasi-TE mode



(b) Group index for the quasi-TE mode

Fig. 3: Effective and Group index variation with wavelength

$E_i/\sqrt{2}$ and intensity into $I_i/2$ in both branches. Let the waveguides in the two branches have lengths L_1 and L_2 with a path difference $\Delta L = L_1 - L_2$. The path difference introduces a phase shift $e^{i\phi}$ where $\phi = k\Delta L$ and wavevector $k = \beta + i\alpha/2$. Here $\beta = 2\pi n_{\text{eff}}(\lambda)/\lambda$ is the propagation constant and is dependent on the refractive index of the medium and wavelength of the input field. On the other hand, $\alpha/2$ represents waveguide losses. Without any loss of generality, we can model this phase shift in only one branch such that the output field will be $E_i e^{i\phi}/\sqrt{2}$. In the lossless case with $\alpha = 0$, the second Y-branch combiner will then combine the two fields such that

$$E_{\text{out}} = \frac{1}{\sqrt{2}} \left[\frac{E_i}{\sqrt{2}} e^{i\phi} + \frac{E_i}{\sqrt{2}} \right] = E_i e^{i\phi/2} \cos(\beta\Delta L/2)$$

Note, $e^{i\phi/2}$ is just a global phase. Since in most practical experiments we measure the power or intensity which is proportional to $|E|^2$, the global phase does not contribute. Thus, the output intensity of the Y-branch combiner will be

$$I_{\text{out}} \propto |E_{\text{out}}|^2 = I_i \cos^2(\beta(\lambda)\Delta L/2)$$

The transfer function—or the ratio of the output and input intensities—of the MZI is then

$$T_{\text{MZI}}(\lambda) = I_{\text{out}}/I_i = \cos^2(\beta(\lambda)\Delta L/2)$$

We can also calculate a parameter called the free-spectral range (FSR). The FSR is an important quantity which determines the separation between the peaks of the interferometer.

Consider a phase shift $\delta = \beta\Delta L$. Since we expect the phase to change by 2π between successive peaks,

$$\Delta\delta = \delta_m - \delta_{m+1} = (\beta_m - \beta_{m+1})\Delta L = 2\pi$$

or

$$\Delta\beta = \beta_m - \beta_{m+1} = 2\pi/\Delta L$$

For small variations in β ,

$$\Delta\beta = \beta_m - \beta_{m+1} \approx -\frac{d\beta}{d\lambda}\Delta\lambda$$

Recall, $\beta = 2\pi n(\lambda)/\lambda$. This gives us

$$\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2} \left(n - \lambda \frac{dn}{d\lambda} \right)$$

Using the above equations, the FSR can then finally be calculated as

$$\Delta\lambda \approx \left| -\frac{2\pi}{\Delta L} \left(\frac{d\beta}{d\lambda} \right)^{-1} \right| = \frac{\lambda^2}{\Delta L} \left(n - \lambda \frac{dn}{d\lambda} \right)^{-1}$$

Note, the term in the brackets is nothing but the group index $n_g = (n - \lambda \frac{dn}{d\lambda})$. Thus, the FSR is finally given as $\Delta\lambda = \lambda^2/n_g\Delta L$.

Lastly, it is interesting to note that in many analyses, it is important to measure the FSR in the frequency space. To convert the FSR from wavelength to frequency is a matter of fact noting that

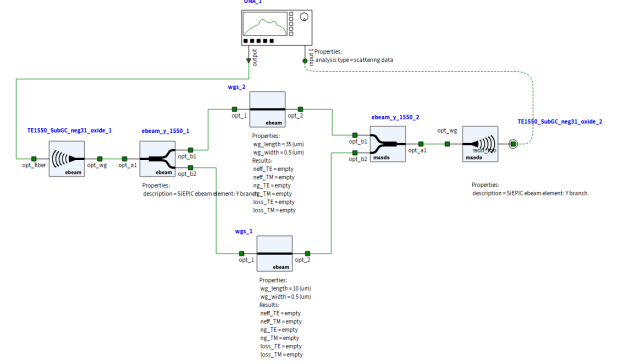
$$\Delta\nu = \left| -\frac{c}{\lambda^2} \Delta\lambda \right| = \frac{c}{n_g\Delta L}$$

C. Modeling MZI

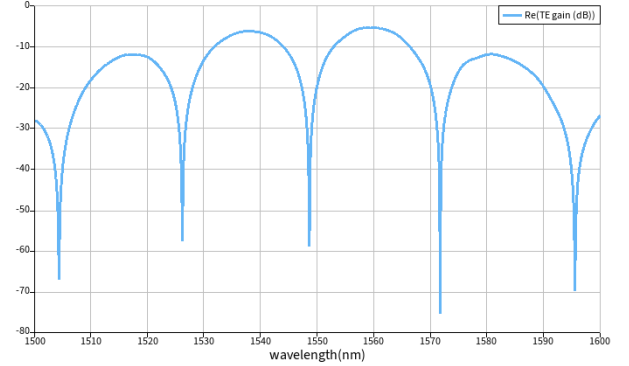
We will perform our simulations of an MZI in Lumerical INTERCONNECT. For this purpose, we will use the EBeam_v2023_02_23.cml compact model library as provided by the edX Phot1X course providers. We install the library in Lumerical INTERCONNECT and use the design kit for our components. We first add two grating couplers ebeam_gc_te1550 for input and output. Then two Y-branches ebeam_y_1550, again one each for input and output. Lastly, we add two strip waveguides ebeam_wg_strip_1550, one for each branch of the MZI. The waveguides have a default width of 500 nm. Let the length of one waveguide be $L_1 = 10 \mu\text{m}$. Depending on the path length difference ΔL , we will vary the length of the other waveguide. We use lengths $L_2 = 35 \mu\text{m}, 60 \mu\text{m}, 110 \mu\text{m}, 210 \mu\text{m},$ and $260 \mu\text{m}$ corresponding to $\Delta L = 25 \mu\text{m}, 50 \mu\text{m}, 100 \mu\text{m}, 200 \mu\text{m},$ and $250 \mu\text{m}$ for our purposes. The outputs were observed on an Optical Network Analyzer. Figs. 4 (a) and (b) show the setup developed in Lumerical INTERCONNECT and the gain spectrum for the case with $\Delta L = 25 \mu\text{m}$. As can be observed from the simulation, the FSR between peaks near 1550 nm and the subsequent peak is 23.12 nm as opposed to 22.84 nm calculated using the formula for $\Delta\lambda$ derived in the last section. The simulated and calculated values are tabulated in Table 1.

TABLE I: FSR for different ΔL

ΔL	Simulated FSR	Calculated FSR
25 μm	23.12 nm	22.84 nm
50 μm	11.39 nm	11.33 nm
100 μm	5.74 nm	5.73 nm
200 μm	2.87 nm	2.86 nm
250 μm	2.29 nm	2.28 nm



(a) MZI setup developed in Lumerical INTERCONNECT for $\Delta L = 25 \mu\text{m}$



(b) Gain spectrum corresponding to the above setup for $\Delta L = 25 \mu\text{m}$

Fig. 4: Setup and spectrum for $\Delta L = 25 \mu\text{m}$

D. KLayout

The GDS layout can be seen in Fig. 5. It contains six MZIs with path length differences $\Delta L = 25 \mu\text{m}, 50 \mu\text{m}, 100 \mu\text{m}, 200 \mu\text{m},$ and $250 \mu\text{m}$. Additionally, it also contains two ring resonators with radii $10 \mu\text{m}$, one made of silicon, and the other made of silicon nitride.

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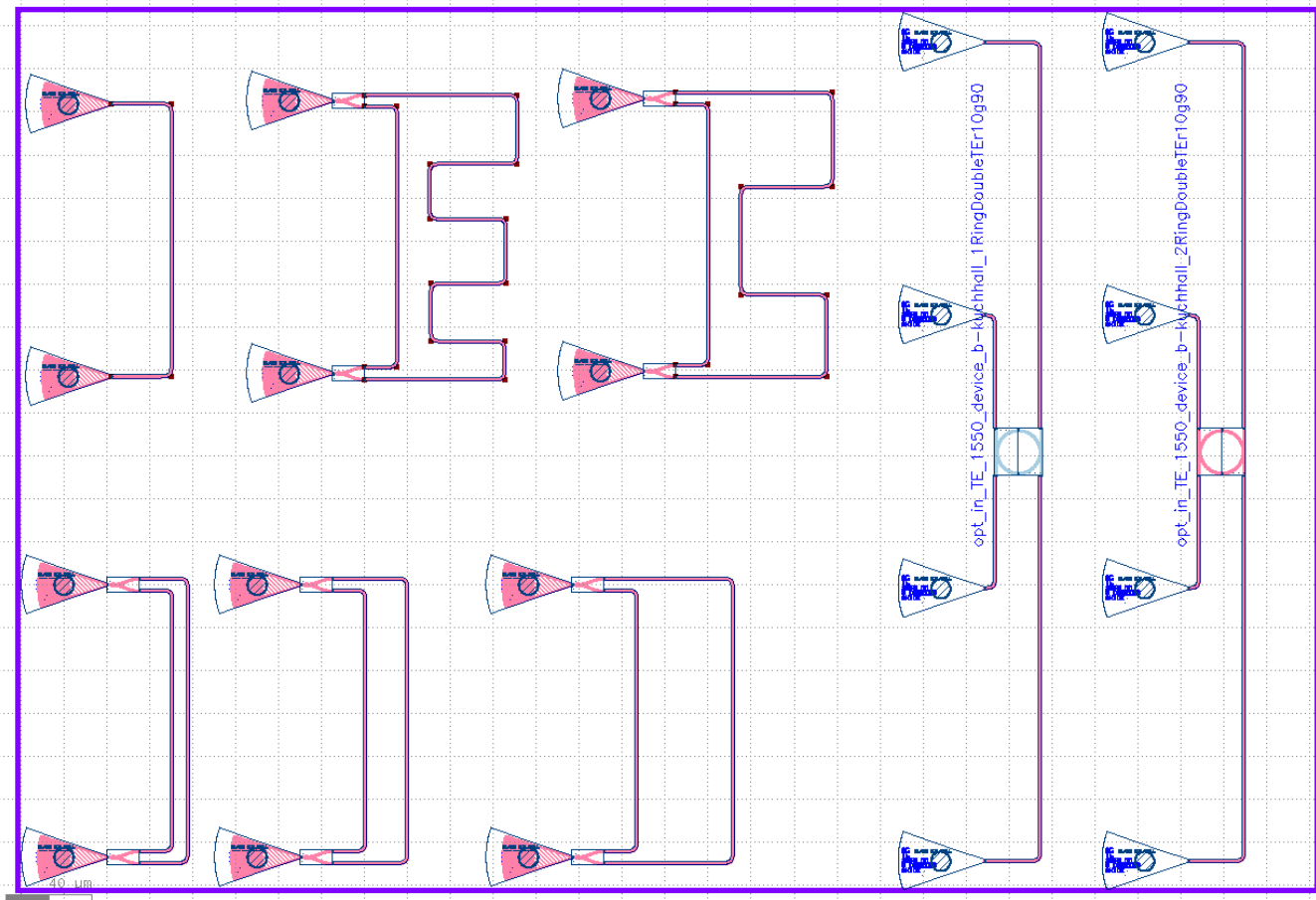


Fig. 5: GDS Layout in KLayout